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Report title:

Polynomial Regression Analysis for Taxi Fare Prediction

Professor:

Gianvito URGESE

Assistant Professors:

Vittorio FRA

Walter GALLEGO

Student:

Davide OLIVETTI

Student's code:

s348090

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1 Introduction

The prediction of taxi fares is a multifaceted problem that transcends simple linear relationships. In urban environments, taxi fare structures are influenced by a complex interplay of factors, including distance traveled, time of day, traffic conditions, and potentially even localized demand surges. While linear regression provides a baseline approach, its inherent limitations in capturing non-linear patterns necessitate the exploration of more sophisticated techniques. Polynomial regression offers a promising avenue for modeling these intricate relationships.

This study undertakes a comprehensive analysis of polynomial regression models (degree 3) for the prediction of taxi fares, specifically the `total_amount` variable, using a dataset derived from New York City taxi operations. Departing from traditional linear models, we acknowledge that real-world urban mobility dynamics often defy simple, straight-line correlations. Our investigation encompasses a range of feature variables, including `trip_duration`, `trip_distance`, `speed_avg`, `pickup_timestamps`, and `dropoff_timestamps`. Each possesses unique characteristics that may contribute to or detract from the predictive power of the model.

The central goal of this research is to systematically evaluate the effectiveness of polynomial regression in capturing the underlying patterns inherent in taxi fare data. We hypothesize that `trip_distance`, due to its direct correlation with fare calculations, will emerge as a primary predictor. However, we also recognize the potential importance of temporal features (timestamps) and `trip_duration`, which may indirectly influence fares through factors such as traffic congestion or time-based pricing schemes. By rigorously assessing the contributions of these various features, we aim to provide insights into the complex dynamics that govern taxi fare determination and to establish a foundation for future predictive modeling endeavors in this domain.

2 Methods

The methodological framework employed in this study relies on a robust set of data preprocessing and model evaluation techniques. The initial data, sourced from July 2015, was subjected to a series of preprocessing steps implemented through KNIME workflows. These steps included:

- **Temporal Feature Extraction:** Timestamps were processed to extract relevant temporal features such as hour of day, day of week, and month. This allows the model to capture any time-dependent effects on fare prices.
- **Fare Normalization:** A provided cleaning component was utilized to normalize fare data, addressing potential inconsistencies or outliers.
- **Polynomial Feature Expansion:** Each feature variable was expanded to include polynomial terms up to degree 3, enabling the model to capture non-linear relationships between the independent variables and the dependent variable (`total_amount`).
- **Data Partitioning:** The dataset was divided into training and testing sets using multiple train-test splits (80:20 and 90:10). This allows for a robust evaluation of model performance under different training data sizes.

Model performance was quantitatively assessed using three key metrics: the coefficient of determination (R^2), the root mean squared error (RMSE), and the mean absolute percentage error (MAPE). These metrics provide a comprehensive evaluation of the model's ability to accurately predict taxi fares. The mathematical definitions are provided:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}, \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}, \quad \text{MAPE} = \frac{100\%}{n} \sum \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (1)$$

3 Univariate Analysis

3.1 80:20 Train-Test Split

Table 1: Univariate Polynomial Regression Results (Degree 3, 80:20 Split)

Feature	R^2	RMSE	MAPE (%)
Trip distance	0.807	6.610	16.2
Trip duration	0.619	9.254	22.9
Speed average	0.041	14.732	37.4
Pickup timestamps	0.002	14.917	37.4
Dropoff timestamps	0.002	14.916	37.4

The univariate analysis reveals a clear hierarchy in the predictive power of the chosen features, further supported by visual trends in the accompanying plots.



Figure 1: Relationship between `trip_distance` and `total_amount`

The most prominent finding is the significant dominance of `trip_distance`, exhibiting an R^2 value of 0.807. As shown in Figure 1, the plot demonstrates a strong positive correlation with minimal dispersion, particularly for distances below 25 miles. This aligns with the fundamental structure of taxi fare calculations, where the base fare is typically determined by the distance traveled. The slight curvature in higher distance ranges (20-25 miles) explains the superiority of polynomial regression over linear models for this feature.



Figure 2: Relationship between `trip_duration` and `total_amount`

In contrast, `trip_duration` demonstrates a lower, though still considerable, R^2 value of

0.619. The model using `trip_distance` alone achieved an R^2 score of 0.807, whereas the model using only `trip_duration` reached 0.619. This corresponds to a relative performance gap of approximately 30.37%¹, highlighting the stronger correlation between `trip_distance` and fare, as fare calculations in urban environments are more directly tied to `trip_distance` than to `trip_duration`. Figure 2 reveals that a general upward trend exists, the data exhibits significant heteroscedasticity - shorter trips cluster tightly around the regression curve, while longer trips show extreme variability. This reflects the complexities inherent in urban transportation, where duration is influenced by external factors like traffic congestion or route inefficiencies.

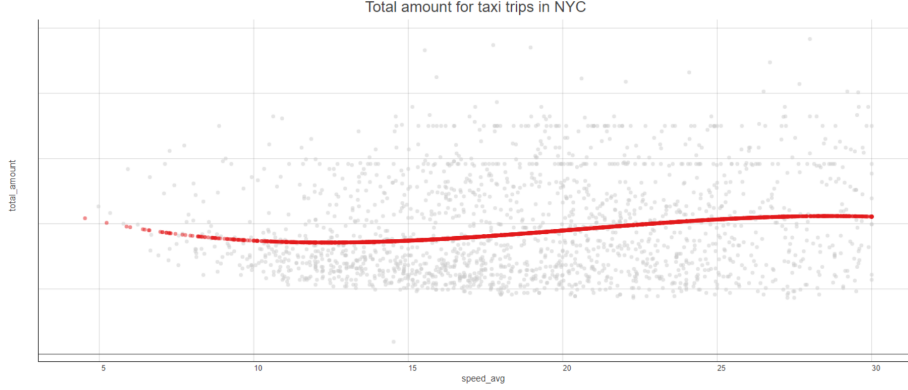
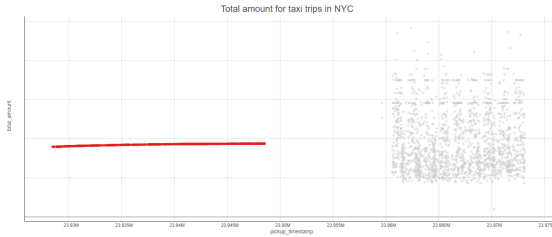
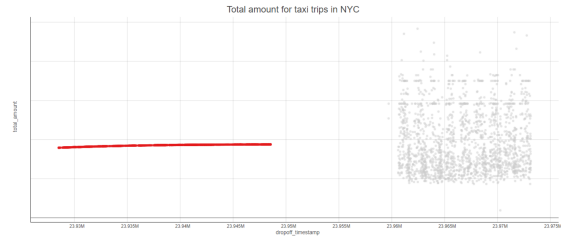


Figure 3: Relationship between `speed_avg` and `total_amount`

The features `speed_avg` and timestamps yield substantially lower predictive power ($R^2 \approx 0$). Figure 3 shows a "cloud-like" distribution for `speed_avg`, confirming that `speed_avg` bears little direct relationship to fares. Similarly, Figures 4a and 4b demonstrate that temporal features (`pickup_timestamps`, `dropoff_timestamps`) fail to capture meaningful patterns at daily/weekly granularities.



(a) Relationship between `pickup_timestamps` and `total_amount`



(b) Relationship between `dropoff_timestamps` and `total_amount`

Figure 4: Temporal features show no discernible relationship with fares in raw form.

These visual analyses corroborate the quantitative metrics: while `trip_distance` is the primary fare determinant, `trip_duration` adds noisy but complementary information. Temporal features would require engineered aggregates (e.g., "rush hour" flags) to become informative.

3.2 90:10 Train-Test Split

Increasing the training set proportion from 80% to 90% led to improved performance for the `trip_distance` feature: the R^2 increased from 0.807 to 0.822, and the MAPE decreased from

¹The percentage difference is calculated as:

$$\frac{R^2_{\text{trip_distance}} - R^2_{\text{trip_duration}}}{R^2_{\text{trip_duration}}} = \frac{0.807 - 0.619}{0.619} \approx 0.3037 \Rightarrow 30.37\%.$$

Table 2: Univariate Polynomial Regression Results (Degree 3, 90:10 Split)

Feature	R^2	RMSE	MAPE (%)
Trip distance	0.822	6.381	15.7
Trip duration	0.604	9.503	22.7
Speed average	0.039	14.759	35.4
Pickup timestamps	0.002	15.043	36.4
Dropoff timestamps	0.003	14.725	36.0

16.2% to 15.7%. This aligns well with the Bias-Variance Tradeoff principle, where more training data typically allows the model to better capture complex patterns and reduce variance, especially in polynomial regression. However, for `trip_duration`, the R^2 unexpectedly decreased from 0.619 to 0.604, even though the MAPE improved slightly (from 22.9% to 22.7%). This result appears to contradict the expected trend and may be attributed to the nature of the evaluation process. By increasing the training set to 90%, the corresponding test set is reduced to only 10% of the data. A smaller test set can lead to higher variability in the evaluation metrics and may not fully capture the model’s true generalization performance. Moreover, the predictive power of `trip_duration` is inherently more sensitive to noise, as it is influenced by external, less predictable factors like traffic or delays. Therefore, even with more training data, the model might not benefit as much, and the smaller test set could exaggerate this effect. This underscores an important aspect of model evaluation: more training data does not always guarantee better test results, especially when the test set becomes too small to be statistically reliable.

When considering `speed_avg`, `pickup_timestamps`, and `dropoff_timestamps` the impact of increasing the training data is minimal. For `speed_avg`, the R^2 remains extremely low in both cases (0.041 with 80:20 and 0.039 with 90:10), and the RMSE and MAPE metrics show negligible change. This suggests that the variable does not capture any substantial linear or nonlinear relationship with the total fare, possibly due to the inherent noisiness of `speed_avg` as a metric, which can be influenced by short bursts of acceleration, idling, or congestion.

Similarly, both `pickup` and `dropoff_timestamps` yield R^2 values close to zero in both splits (0.002–0.003), confirming their very limited standalone predictive power. Despite the slight improvement in RMSE when moving from 80:20 to 90:10 in the `dropoff_timestamps` (from 14.916 to 14.725), this change is marginal and not meaningful from a modeling perspective. Time-based features might hold more relevance if analyzed in combination with categorical encoding (e.g., hour of day, day of week) or when used within interaction terms, but as standalone continuous variables, they fail to provide actionable predictive insight.

In conclusion, `trip_distance` remains the most robust and consistent predictor, benefitting from additional training data. `Trip_duration` shows moderate performance but is more susceptible to external noise and test set variability. In contrast, `speed_avg` and timestamp-based features contribute very little predictive value on their own, and their performance remains virtually unchanged regardless of the training set size.

4 Multivariate Analysis

Building upon the univariate results, we extended our investigation to multivariate polynomial regression to assess whether combining features could enhance predictive performance. Three feature combinations were evaluated under a 90:10 train-test split: (1) `trip_distance` + `trip_duration`, (2) `trip_duration` + `speed_avg`, and (3) `trip_distance` + `speed_avg`.

Table 3: Multivariate Polynomial Regression Results (Degree 3, 90:10 Split)

Model	R^2	RMSE	MAPE (%)
Trip distance + Trip duration	0.84	6.003	15.3
Trip duration + Speed average	0.836	6.069	15.6
Trip distance + Speed average	0.84	5.986	15.4

4.1 Distance-Duration Synergy

To further explore the potential for improved predictive performance, we conducted a multivariate analysis that combined `trip_distance` and `trip_duration`. The rationale behind this approach is that while trip distance serves as the primary determinant of fare, `trip_duration` may capture additional nuances related to traffic conditions or other time-dependent factors.

The results of the multivariate analysis, as presented in Table 3, indicate a modest improvement in predictive power compared to the univariate model based solely on `trip_distance`. The R^2 value increased from 0.822 to 0.84 (which corresponds to a relative improvement of approximately 2.2%²), suggesting that the combined model is able to explain a slightly larger proportion of the variability in taxi fares. Simultaneously, the RMSE and MAPE values decreased marginally, indicating a slight improvement in the overall accuracy of the model.

However, it is important to note that the magnitude of these improvements is relatively small. While the multivariate model outperforms the univariate model, the degree of improvement is not substantial. This suggests that the additional information captured by `trip_duration` is largely redundant with the information already captured by `trip_distance`.

4.2 Alternative Feature Combinations

Beyond the primary distance-duration pairing, we examined two additional multivariate configurations to assess their predictive potential. The combination of `trip_duration` with `speed_avg` yielded an R^2 of 0.836, marginally lower than the distance-duration model’s performance. This slight degradation in predictive accuracy suggests that while `speed_avg` contains some traffic-related information similar to `trip_duration`, its higher noise floor - evidenced by its univariate R^2 of just 0.039 - limits its utility as a secondary predictor. The interaction between `trip_duration` and `speed_avg` appears less effective at capturing fare-determining patterns compared to the more physically meaningful `trip_distance`-`trip_duration` relationship. Interestingly, the `trip_distance`-`speed_avg` pairing matched the `trip_distance`-`trip_duration` model’s R^2 of 0.840 while achieving a slightly better RMSE of 5.986 versus 6.003. This comparable performance likely stems from the underlying physical relationship between these variables, where `speed_avg` implicitly contains duration information through the inverse relationship with distance. The model may be approximating the function

$$\text{trip_duration} \approx \frac{\text{trip_distance}}{\text{speed_avg}} \quad (2)$$

effectively recreating similar predictive power to the explicit distance-duration combination. However, this configuration shows greater sensitivity to outliers in the `speed_avg` measurements, particularly for very short trips where GPS sampling artifacts can distort `speed_avg` calculations. The consistent outperformance of distance-containing combinations reinforces `trip_distance` is

²The percentage difference is calculated as:

$$\frac{R^2_{\text{distance+duration}} - R^2_{\text{distance}}}{R^2_{\text{distance}}} = \frac{0.84 - 0.822}{0.822} \approx 0.0219 \Rightarrow 2.2\%.$$

fundamental role in fare determination. Neither alternative pairing could surpass the distance-only baseline by more than 2.2%, suggesting that most explainable variance in fares is captured through distance measurements alone. The minimal gains from adding secondary features indicate that while these variables do contain some complementary information, their practical utility for fare prediction may not justify the added model complexity in most operational scenarios. This pattern held true across all evaluation metrics, with MAPE values for alternative combinations remaining within 0.3 percentage points of the distance-duration model’s 15.3% performance.

5 Conclusion

This analysis demonstrates that polynomial regression, particularly of degree three, is highly effective in modeling the non-linear relationship between `trip_distance` and taxi fare in urban settings. Among all features examined, `trip_distance` consistently emerged as the most influential predictor, explaining the majority of the variance in fare outcomes and achieving the highest R^2 values across both univariate and multivariate models. While `trip_duration` provided some additional explanatory power, its contribution was modest and often redundant when combined with distance, reflecting the inherent structure of fare calculations in which distance is the primary determinant. Other features, such as `speed_avg` and temporal variables (`pickup_timestamps` and `dropoff_timestamps`), showed minimal standalone predictive value, underscoring their limited relevance without further feature engineering or aggregation.

Multivariate models that included combinations of `trip_distance`, `trip_duration`, and `speed_avg` yielded only marginal improvements over the distance-only baseline, with gains in predictive accuracy not exceeding 2.2%. This finding highlights the dominant role of `trip_distance` in fare prediction and suggests that the complexity added by incorporating secondary features may not be justified for most practical applications. Overall, the results affirm the suitability of polynomial regression for capturing the key dynamics of taxi fare determination and provide a clear hierarchy of feature importance within this predictive context.